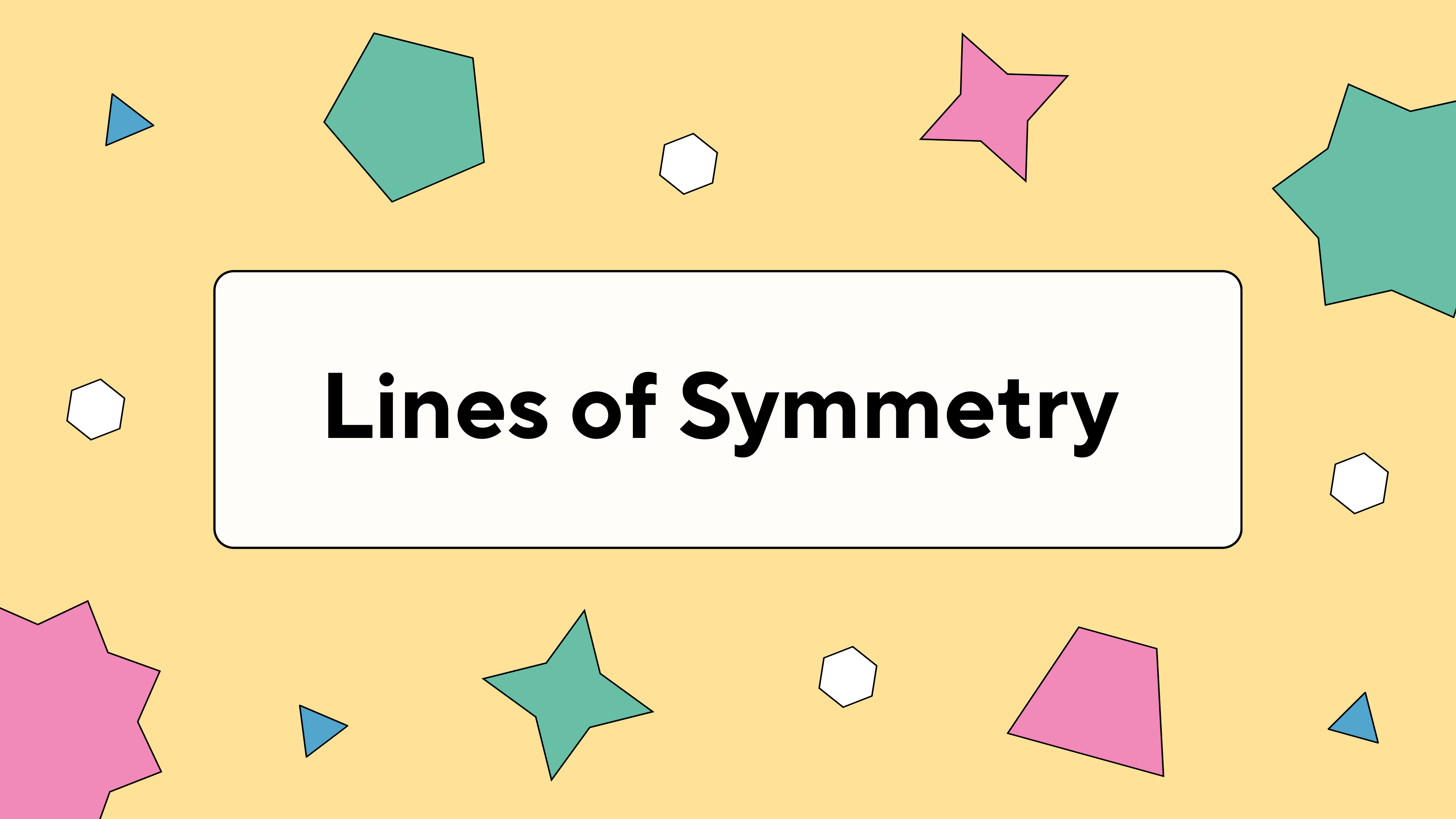
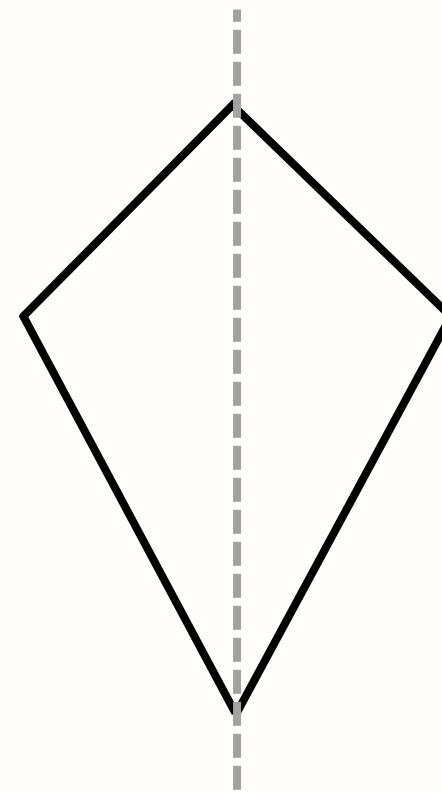
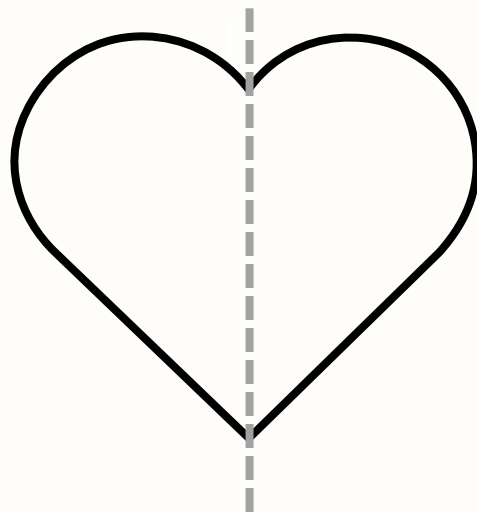
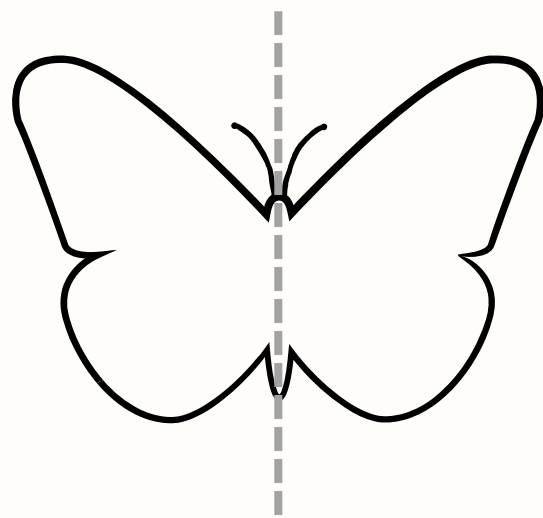


The background is a solid light yellow color. It is decorated with several geometric shapes: a small blue triangle in the top left, a teal pentagon in the top left, a small white hexagon in the top center, a pink five-pointed star in the top right, a large teal five-pointed star on the right edge, a small white hexagon on the left edge, a large pink five-pointed star in the bottom left, a small blue triangle in the bottom left, a teal four-pointed star in the bottom center, a small white hexagon in the bottom center, a pink trapezoid in the bottom right, and a small blue triangle in the bottom right.

Lines of Symmetry

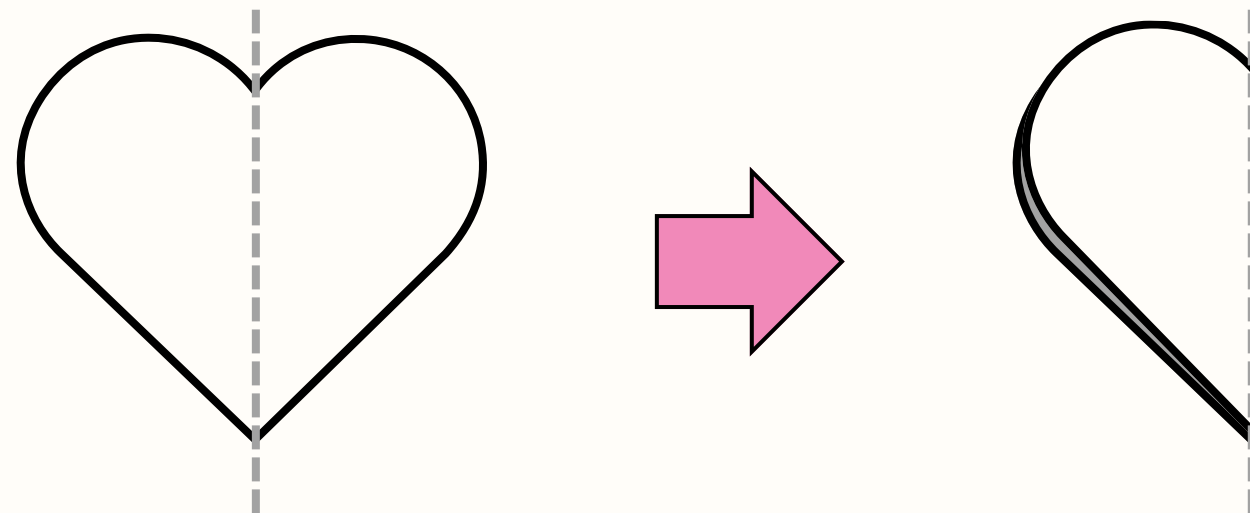
What is Symmetry?

Symmetry is when a shape or object looks exactly the same on both sides.

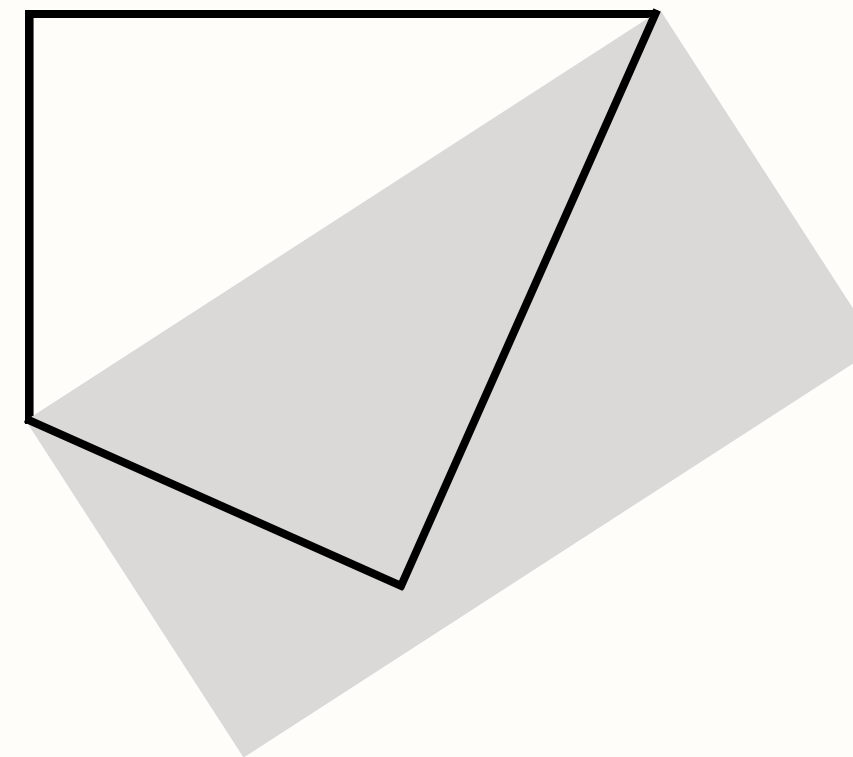
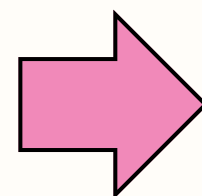
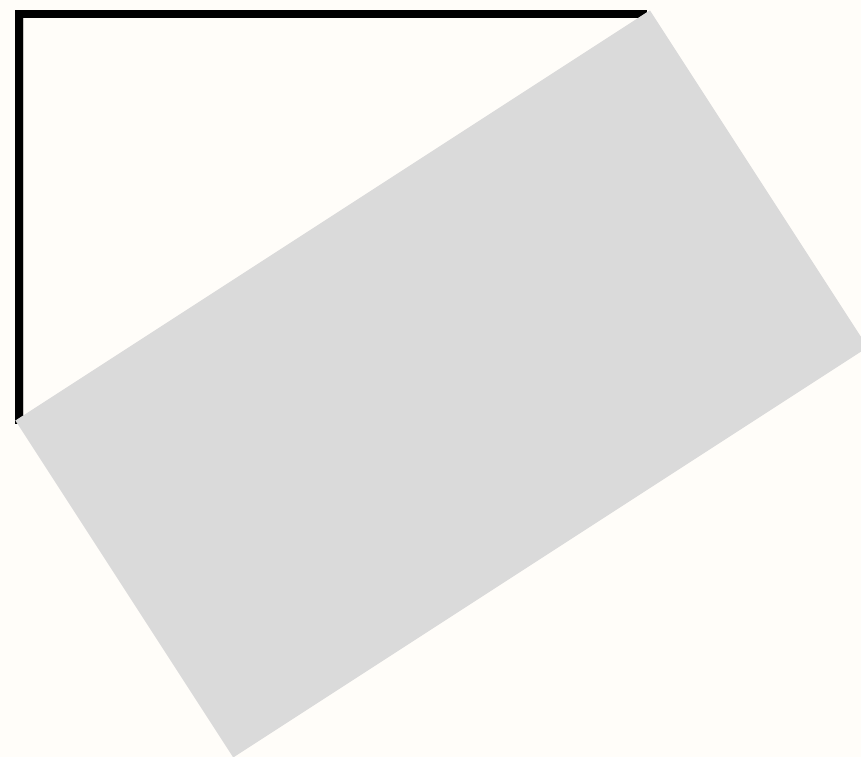
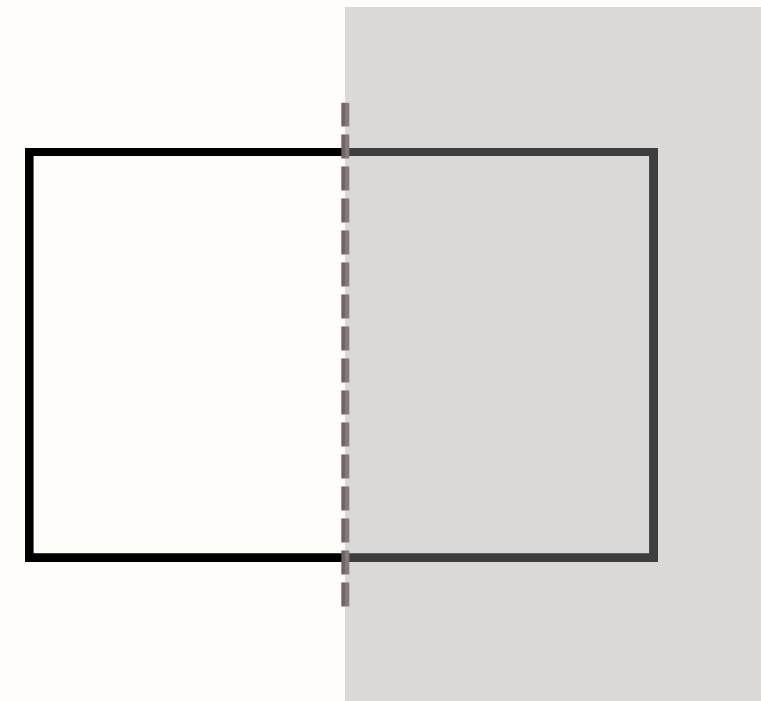
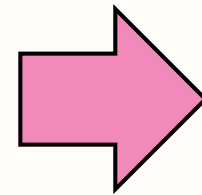
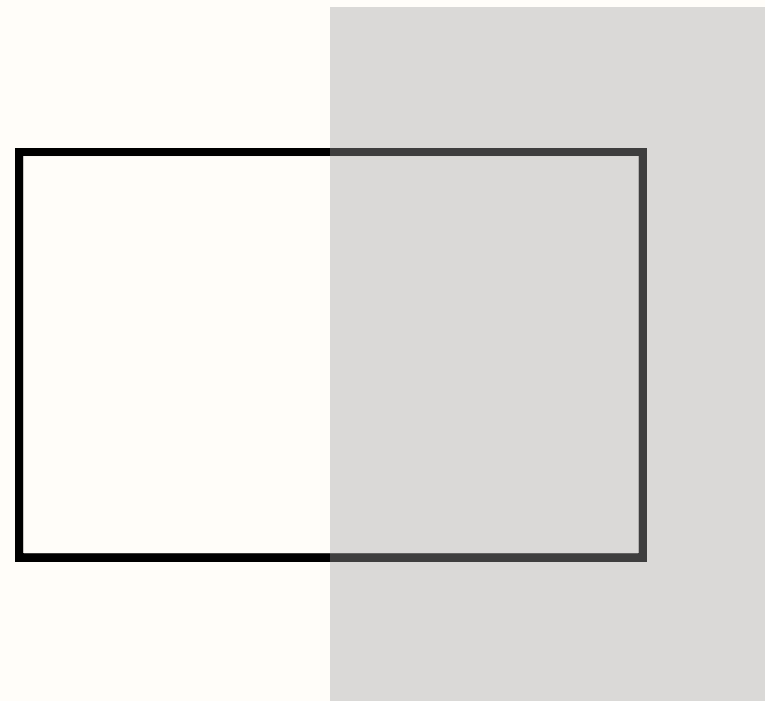


Identifying Lines of Symmetry

A line of symmetry is an imaginary line where you can fold/cut a shape, leaving you with two identical halves.

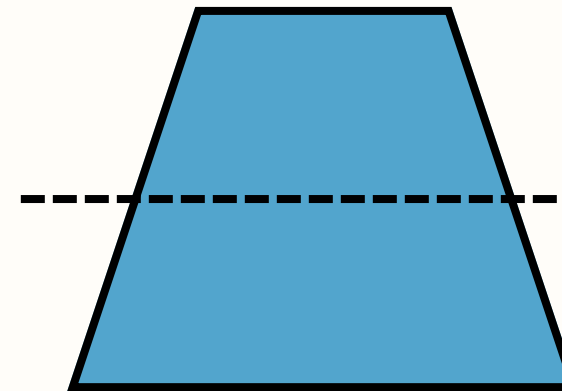
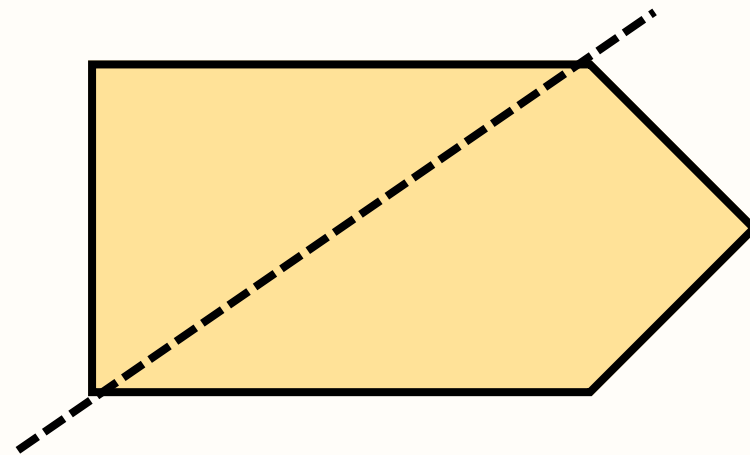
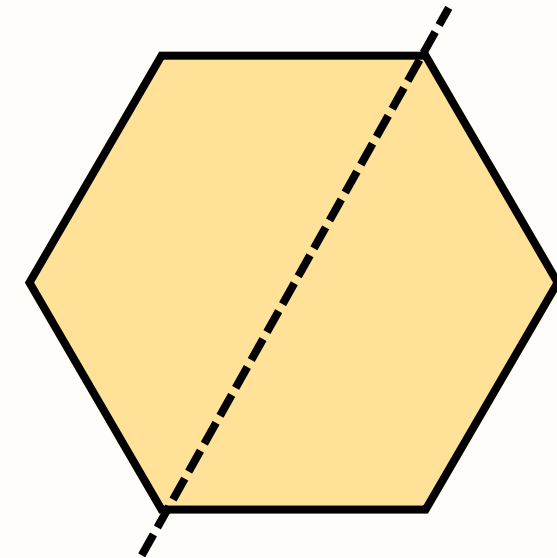
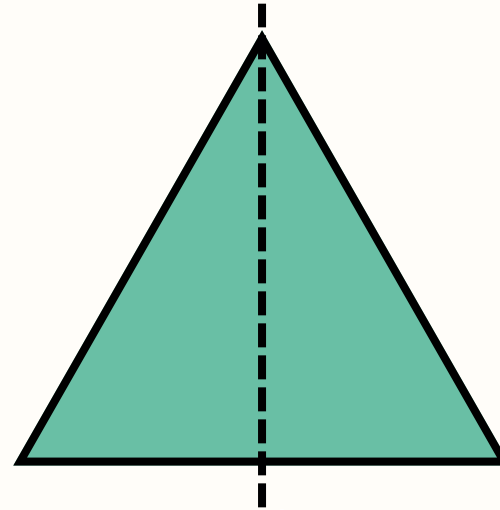
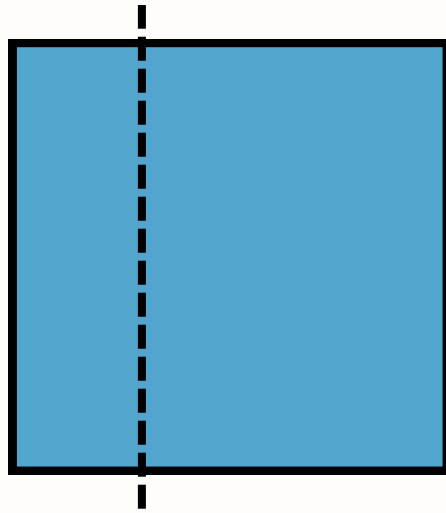


You may also be able to use a mirror to identify lines of symmetry:



*No longer looks
like a rectangle*

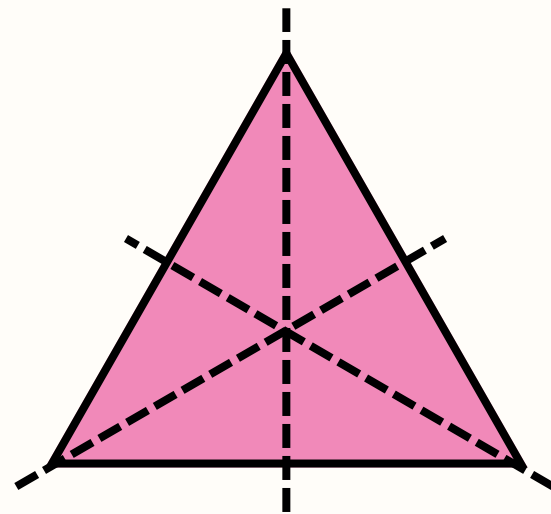
Tick the shapes where the line of symmetry has been identified correctly:



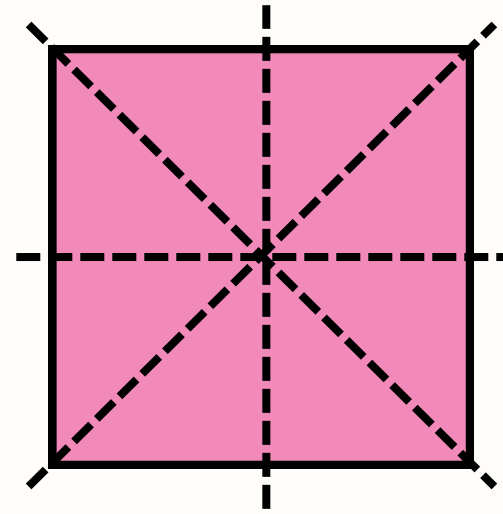
Lines of Symmetry in Regular Shapes

A regular shape has equal side lengths and equal angles.

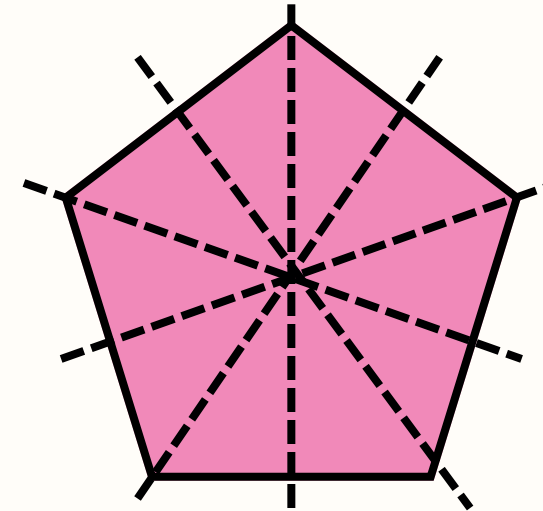
Count along to see how many lines of symmetry the regular shapes have:



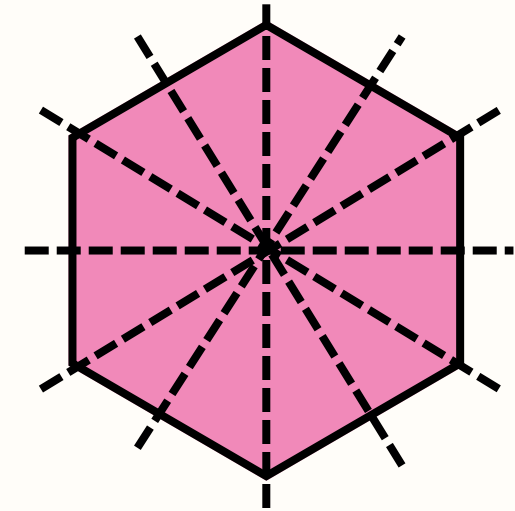
3



4



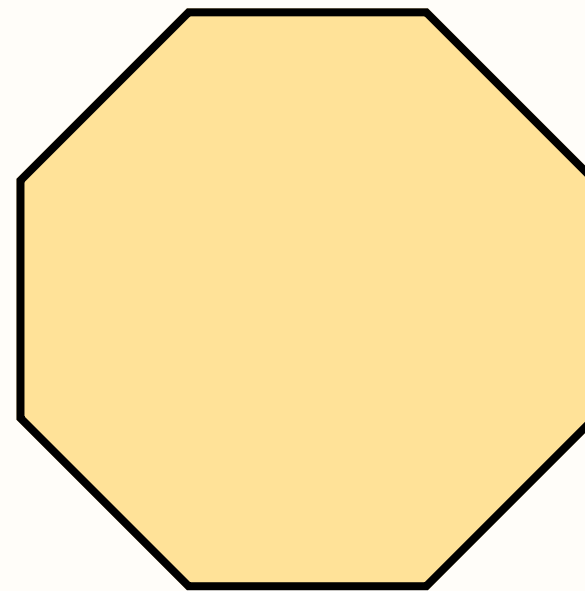
5



6

Did you spot a Pattern?

Can you predict how many lines of symmetry there are in this regular polygon?



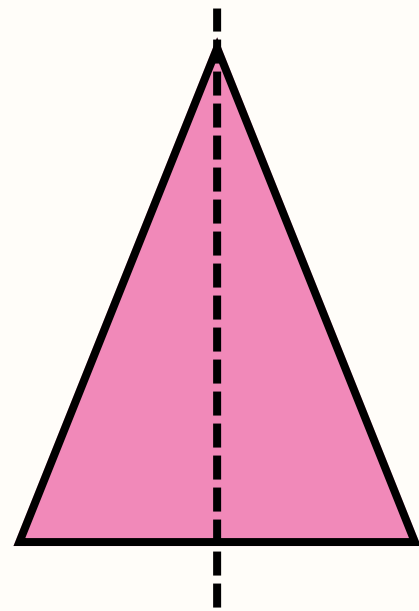
8

Regular polygons have an equal number of sides and lines of symmetry.

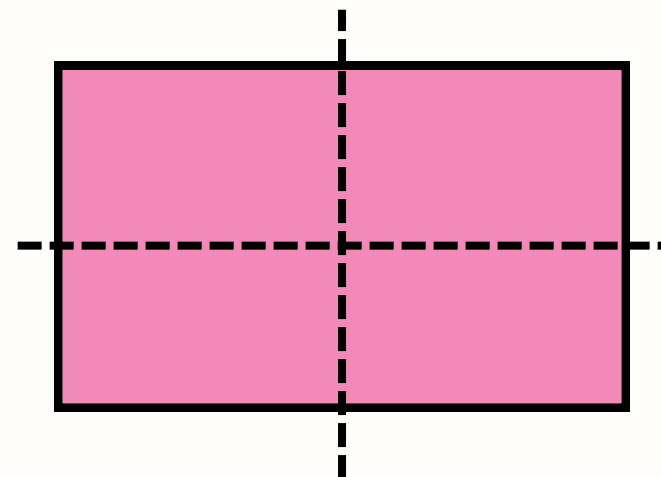
Lines of Symmetry in Irregular Shapes

An irregular shape has different side lengths and angles

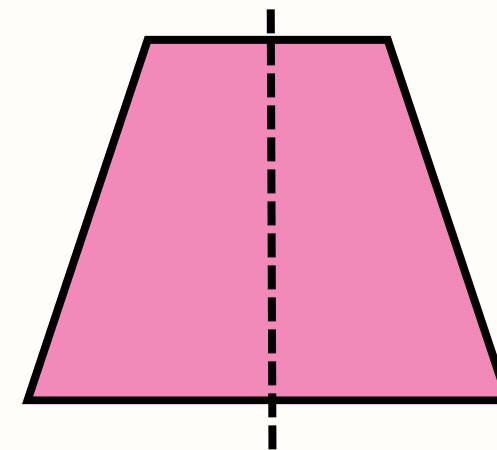
Can you figure out how many lines of symmetry the following shapes have?



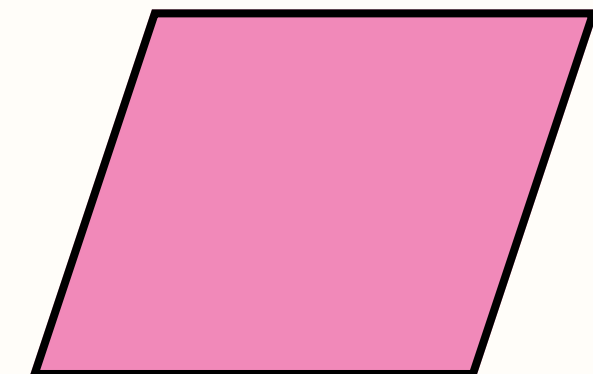
1



2



1



0

Coordinate Planes & Plotting

Introduction to Coordinate Systems



**What are
Coordinates?**



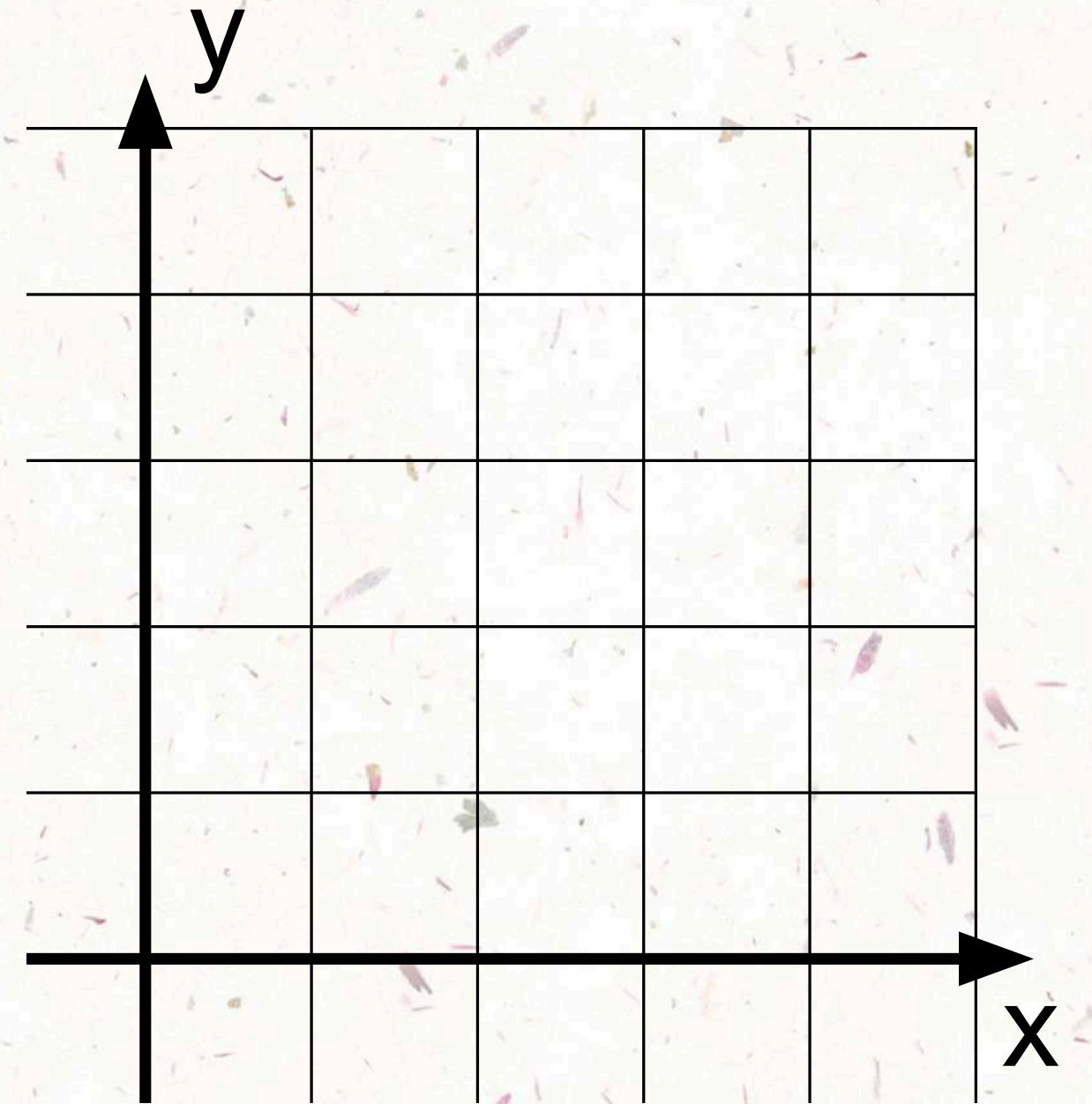
What are Coordinates?

Coordinates are pairs of numbers used to locate points on a plane. This plane is defined by two lines called axes: the x-axis (horizontal) and the y-axis (vertical), which intersect at a point called the origin.



Parts of a Coordinate

Coordinates are pairs of numbers indicating a point's horizontal (x) and vertical (y) positions from the origin, written as (x, y) .



**How do you
Make a
Coordinate?**

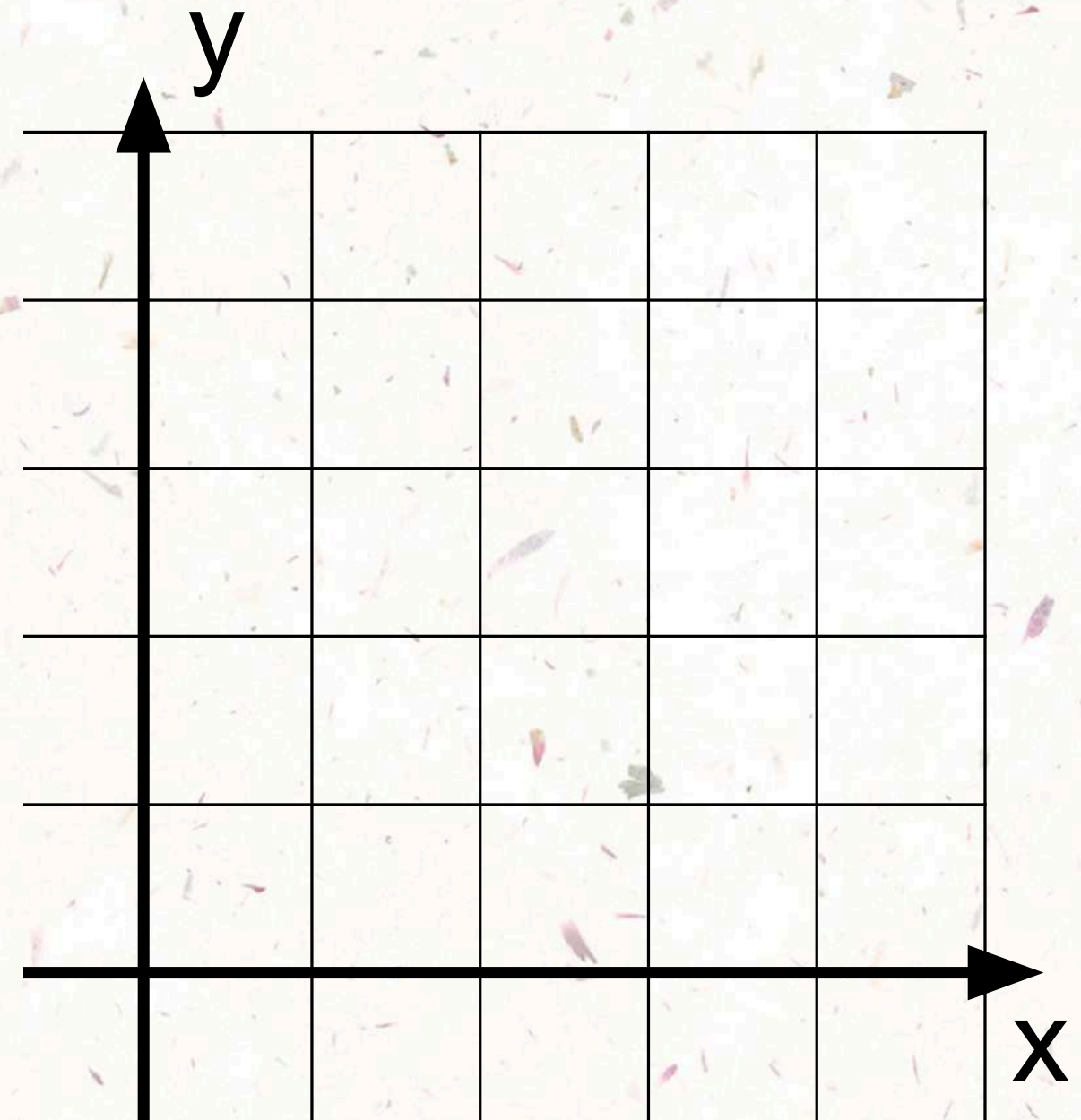


Let's Practice!

Drag the dot to the
following coordinates!



(3,4)

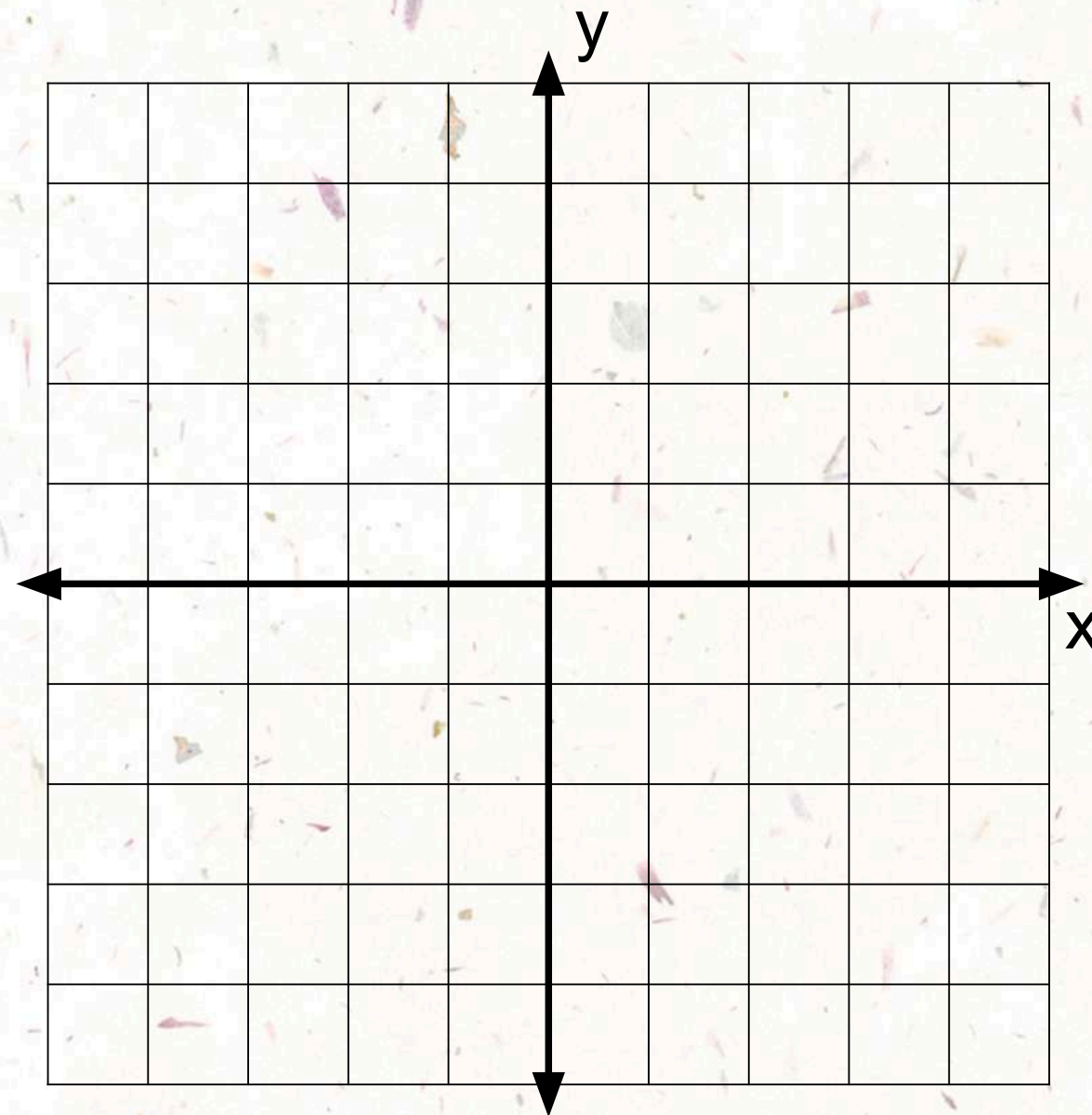


What is a Quadrant Plane?



Quadrant Plane

The coordinate plane is formed by perpendicular x-axis (horizontal) and y-axis (vertical) lines that intersect at the origin.



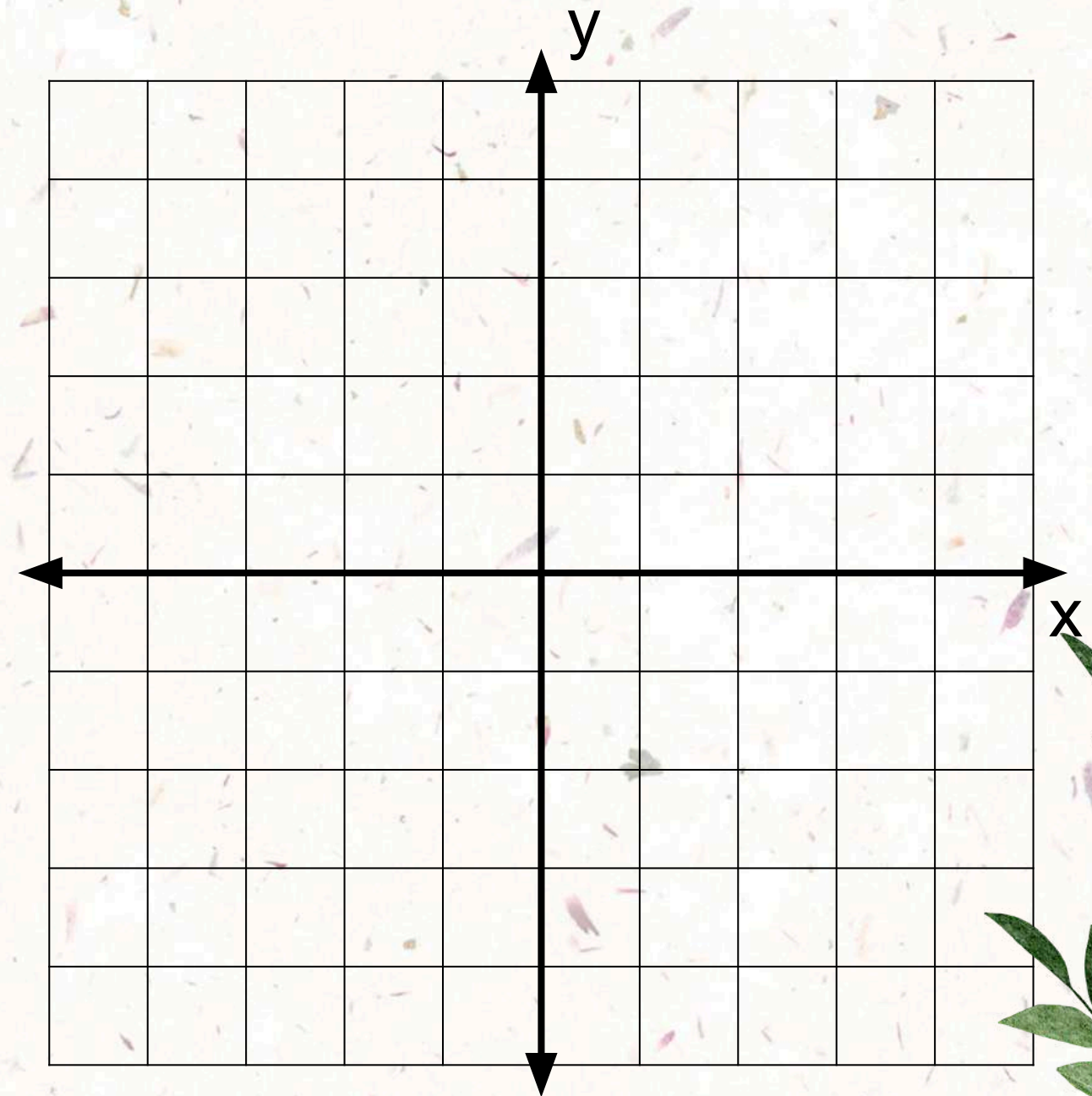
It is divided into four quadrants and is used to graphically represent pairs of numbers as points.

Let's Practice!

Drag the dot to the
following coordinates!



$(-4, 5)$

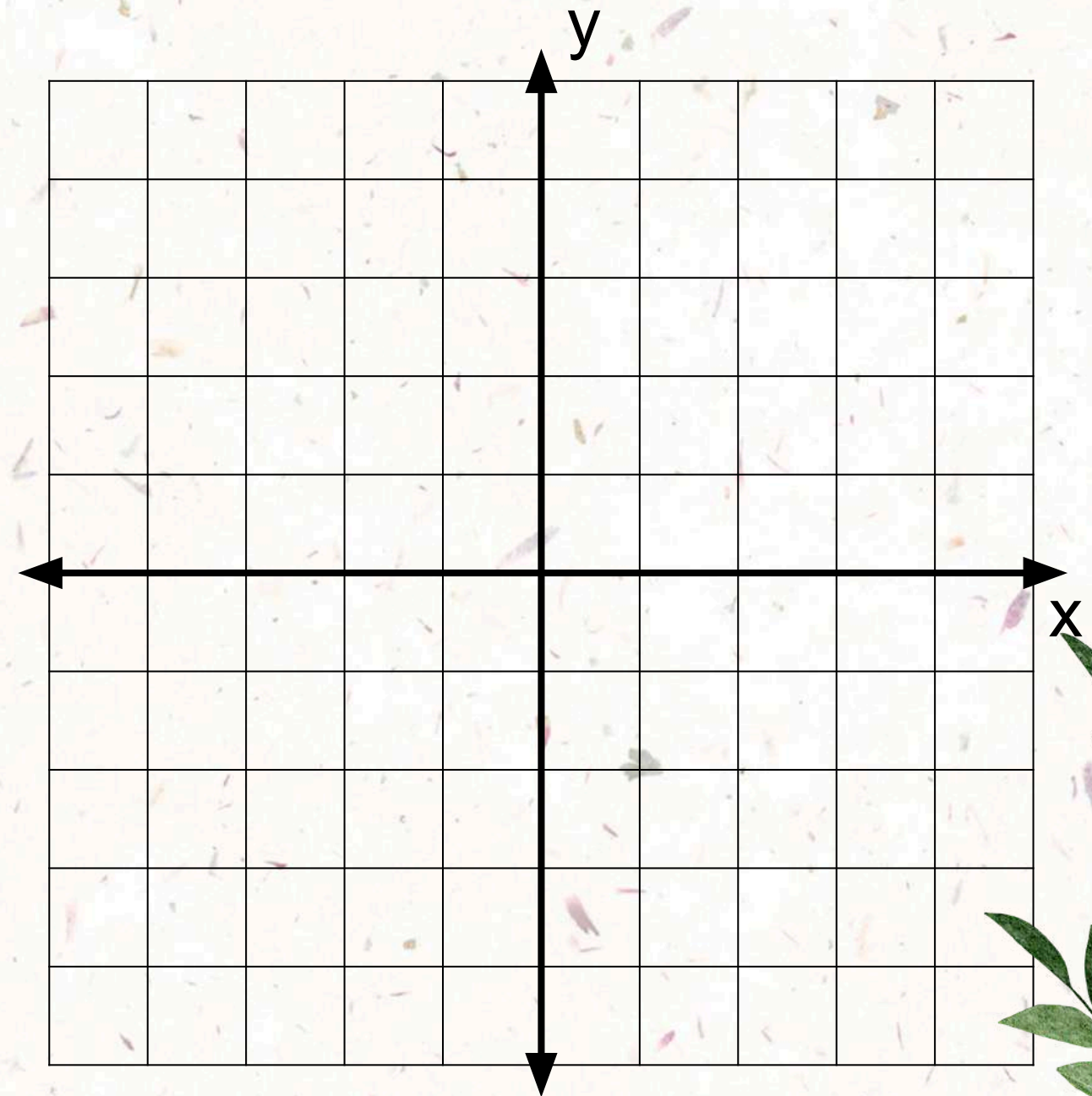


Let's Practice!

Drag the dot to the
following coordinates!



$(2, -3)$

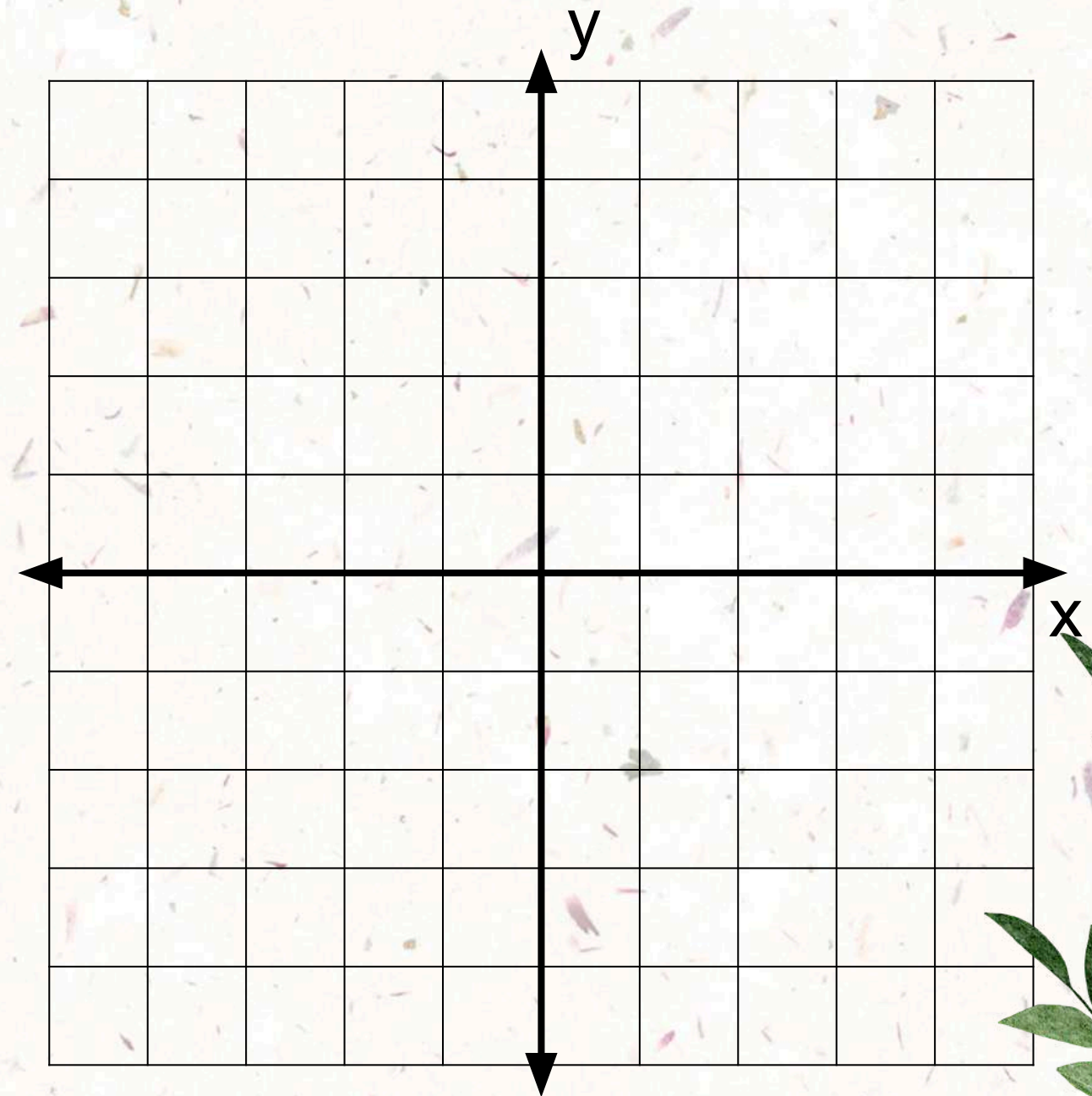


Let's Practice!

Drag the dot to the
following coordinates!



$(-4, 4)$

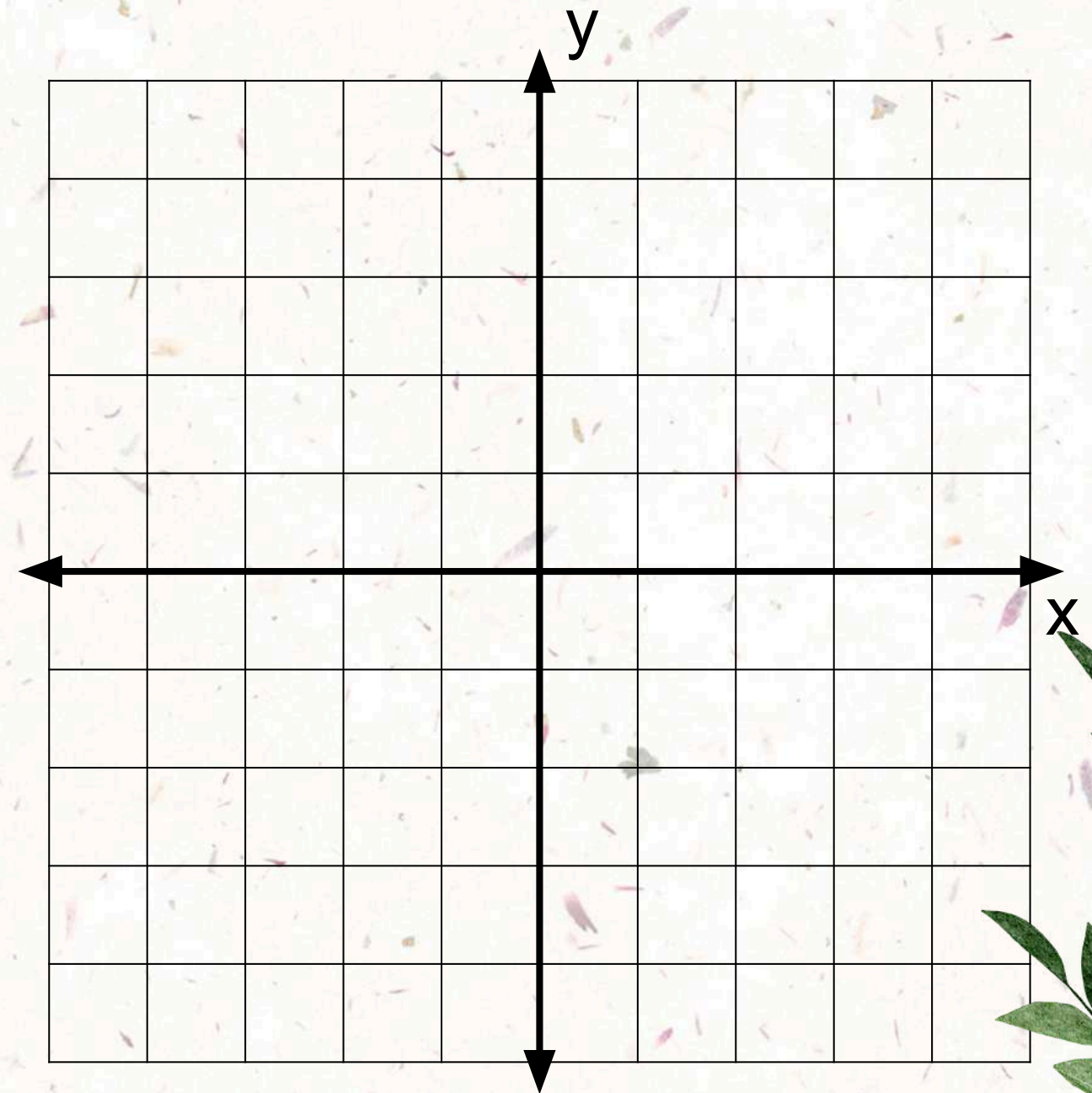


Let's Practice!

Drag the dot to the
following coordinates!



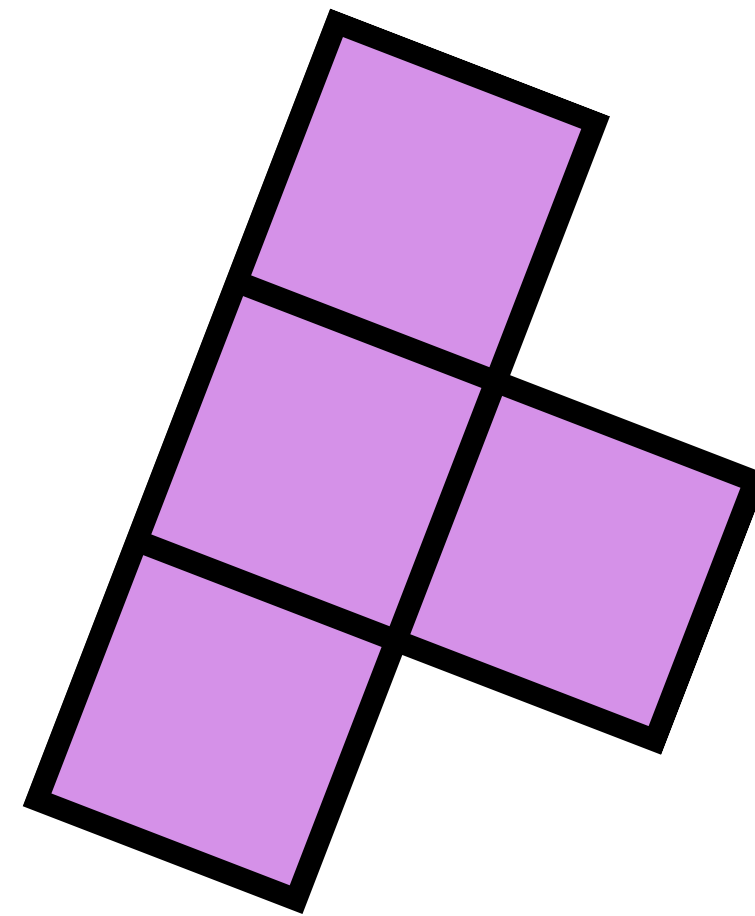
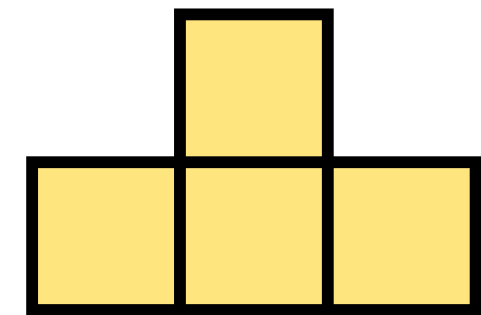
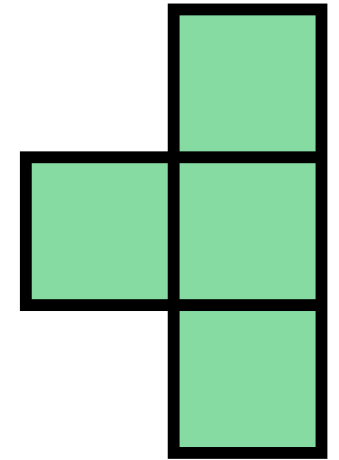
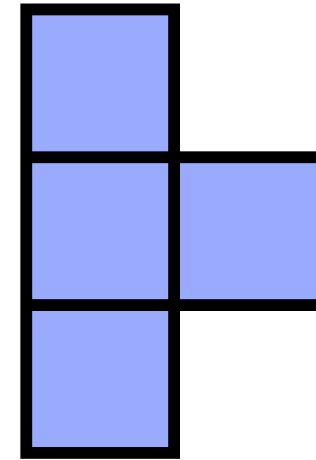
(1,-5)



GEOMETRIC
TRANSFORMATIONS

INTRODUCTION

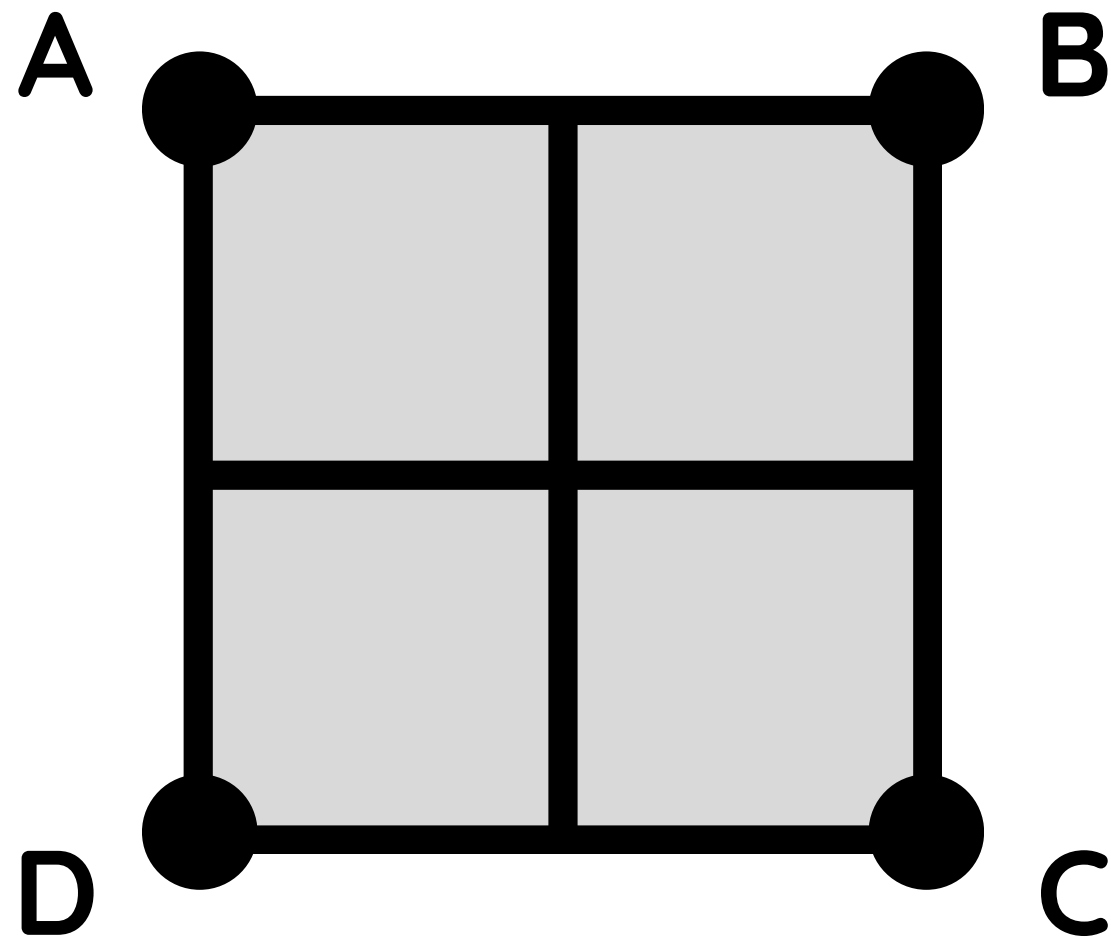
Geometric transformations are operations that change the position, size, or orientation of shapes while preserving their basic properties.



NOTATION

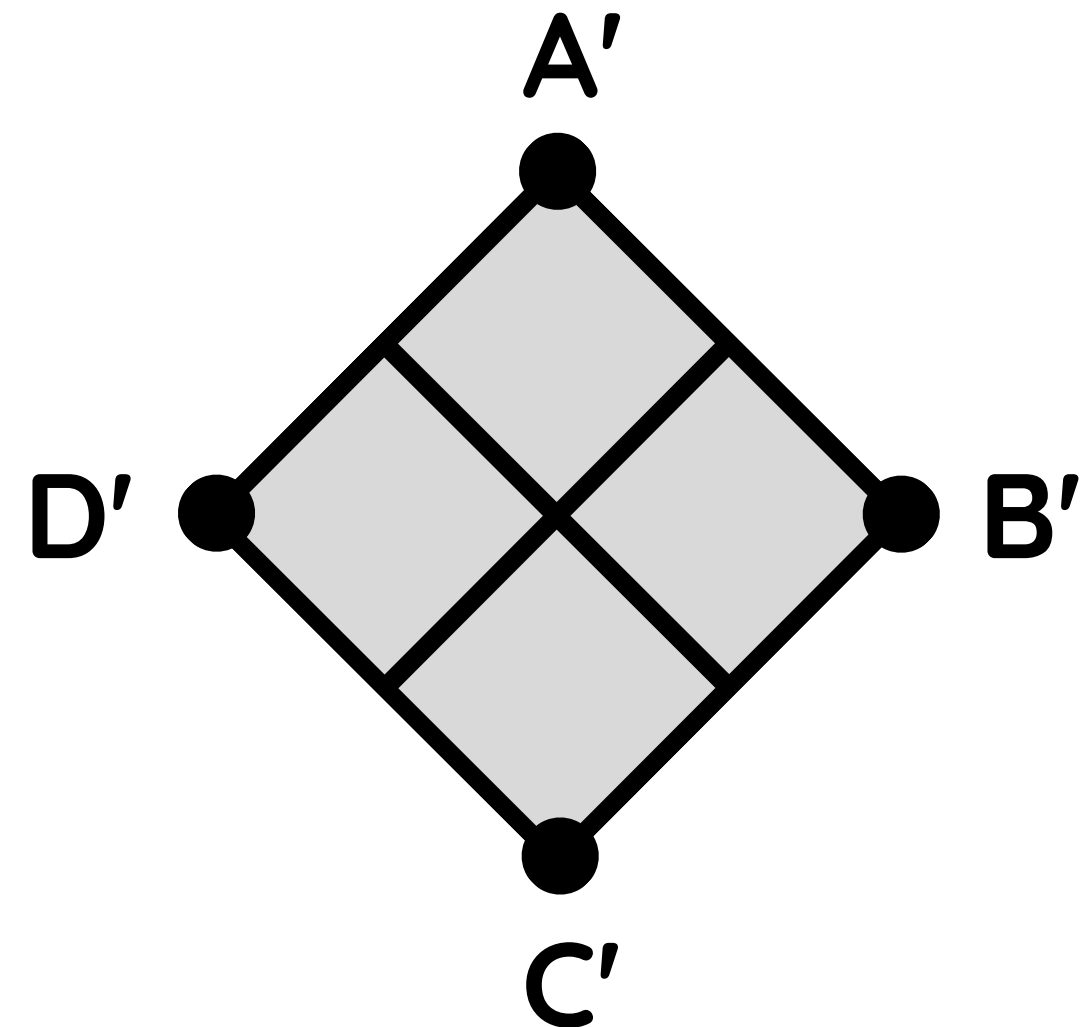
Pre-image

before transformation(s)



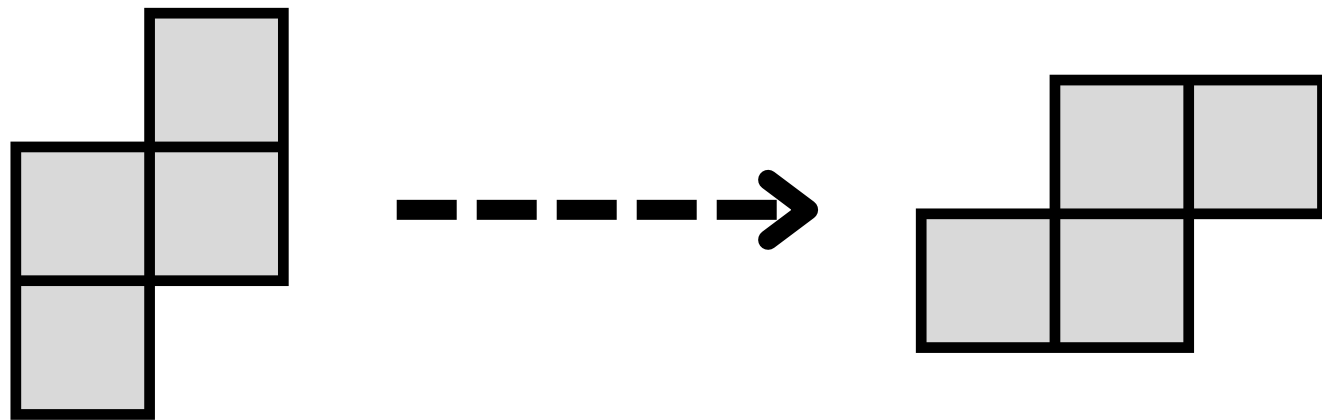
Image

after transformation(s)



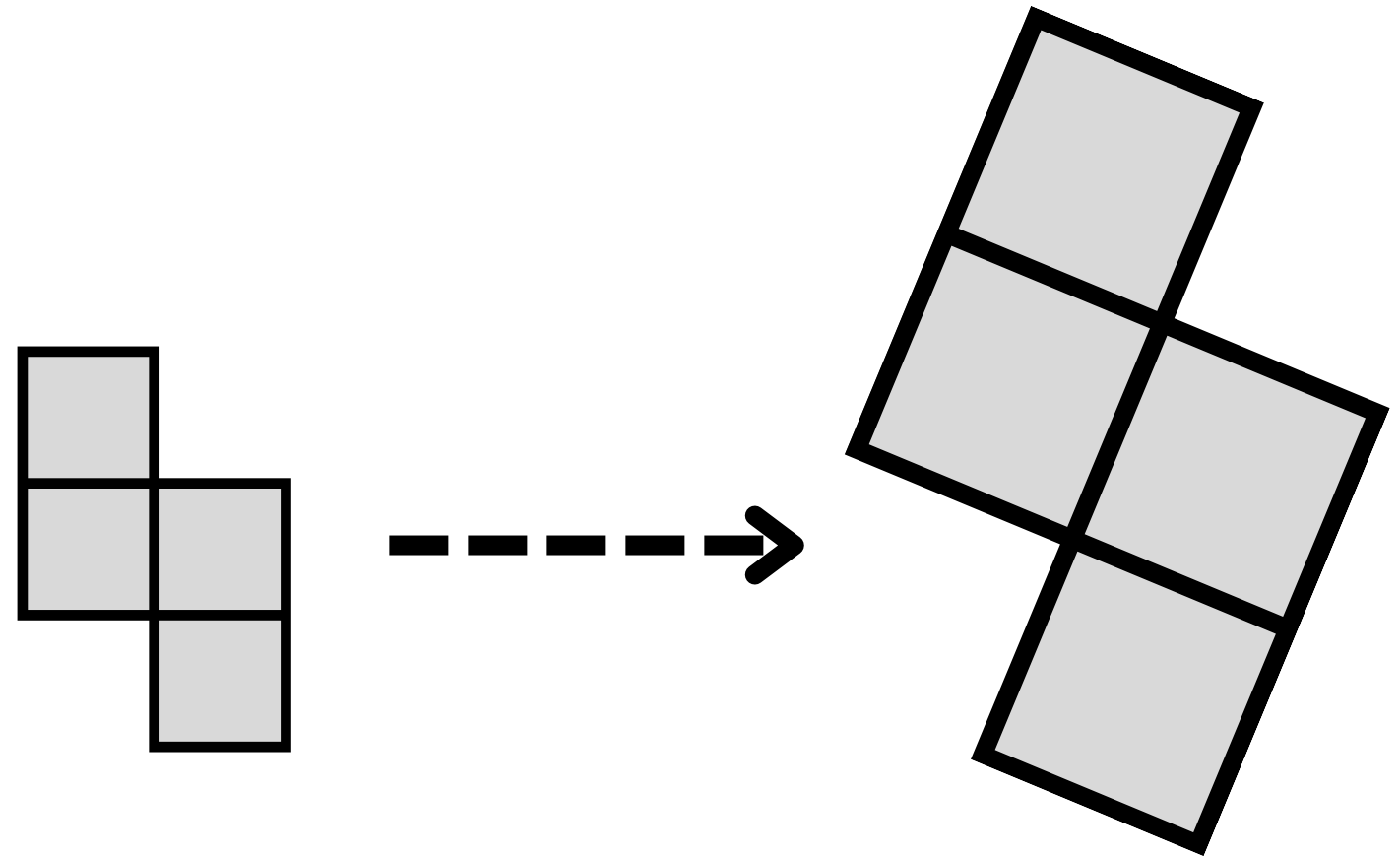
RIGID

- Preserves size and shape
- Also called isometries

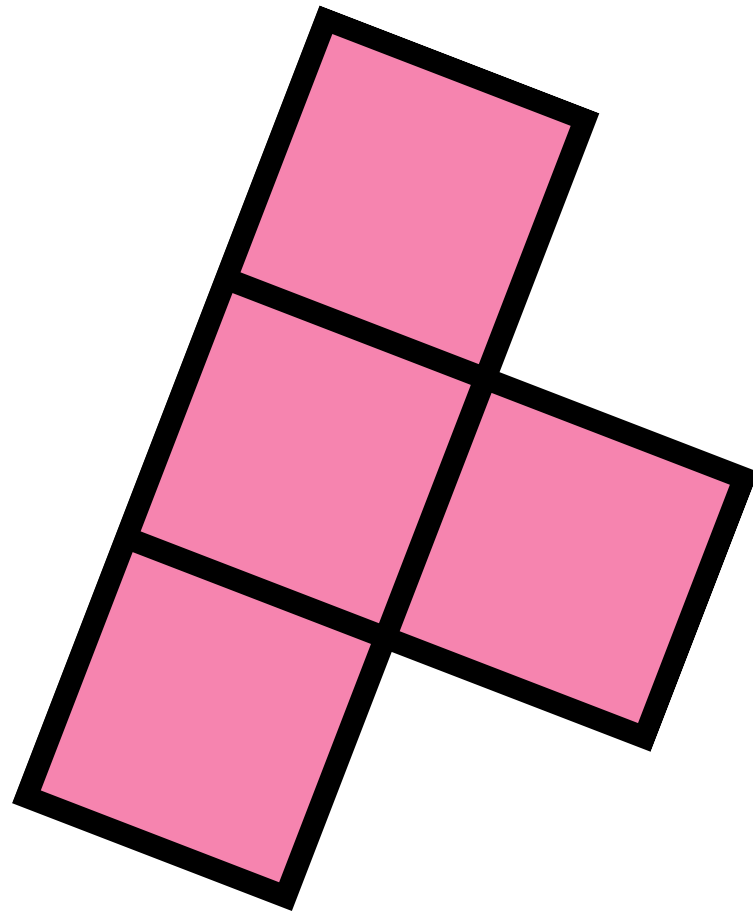
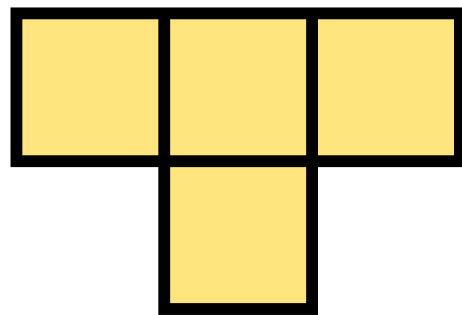
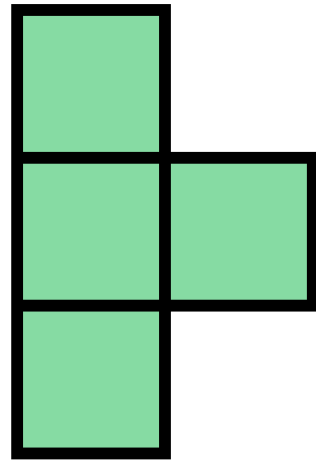
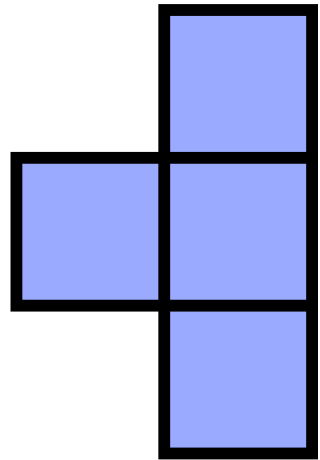


NON-RIGID

- Preserves shape but not size



TYPES OF TRANSFORMATIONS



Rigid

- Translation - "slide"
- Reflection - "flip"
- Rotation - "turn"

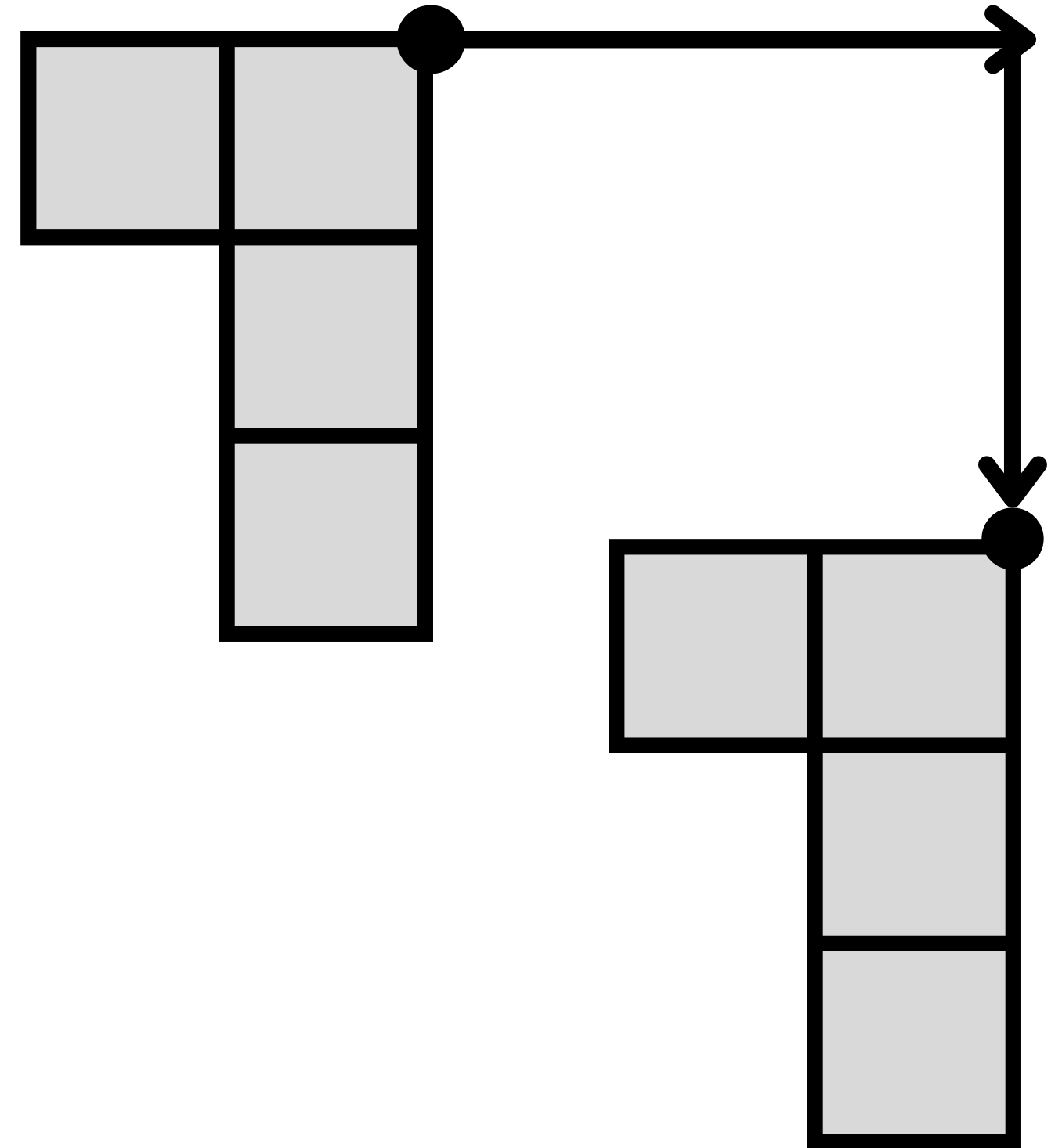
Non - Rigid

- Dilation - "stretch or shrink"

TRANSLATION

A rigid transformation in which a shape is moved vertically, horizontally, or both.

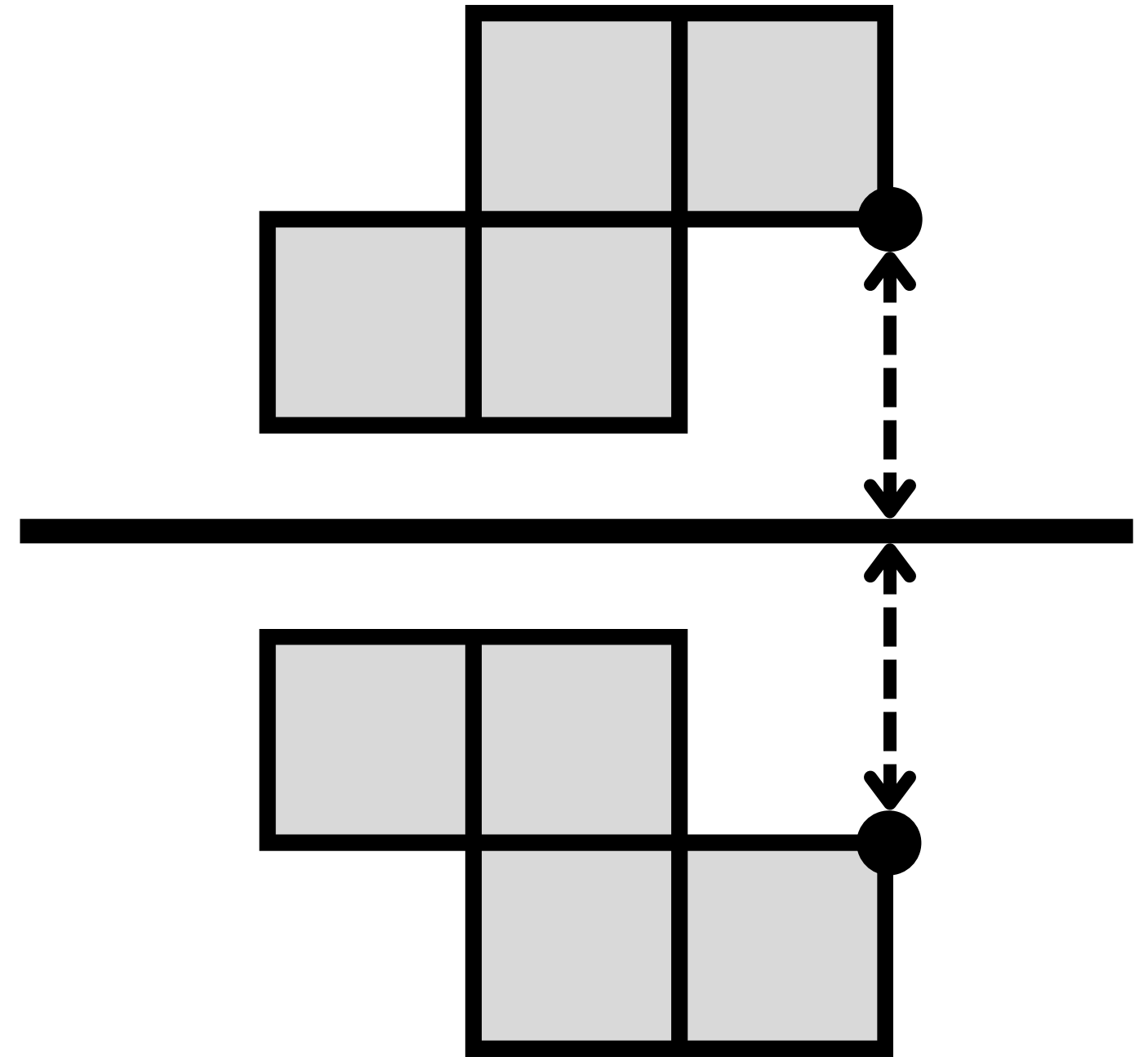
Every point of the pre-image must move the same distance in the same direction.



REFLECTION

A rigid transformation in which an shape is mirrored across a line called the line of reflection.

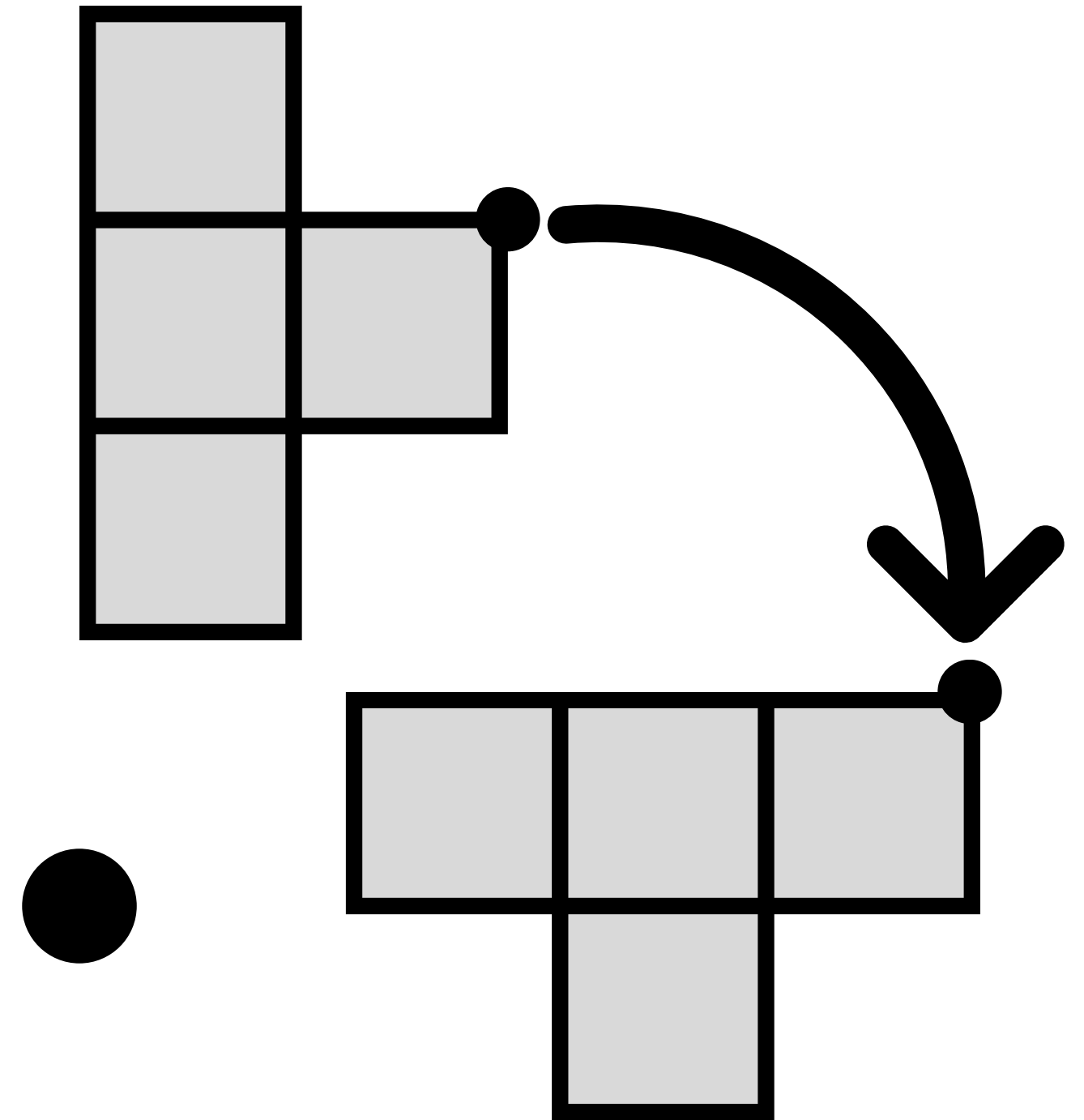
Every point in the image is the same distance from the line of reflection as their corresponding points in the pre-image.



ROTATION

A rigid transformation in which a shape is turned a certain number of degrees around a given point.

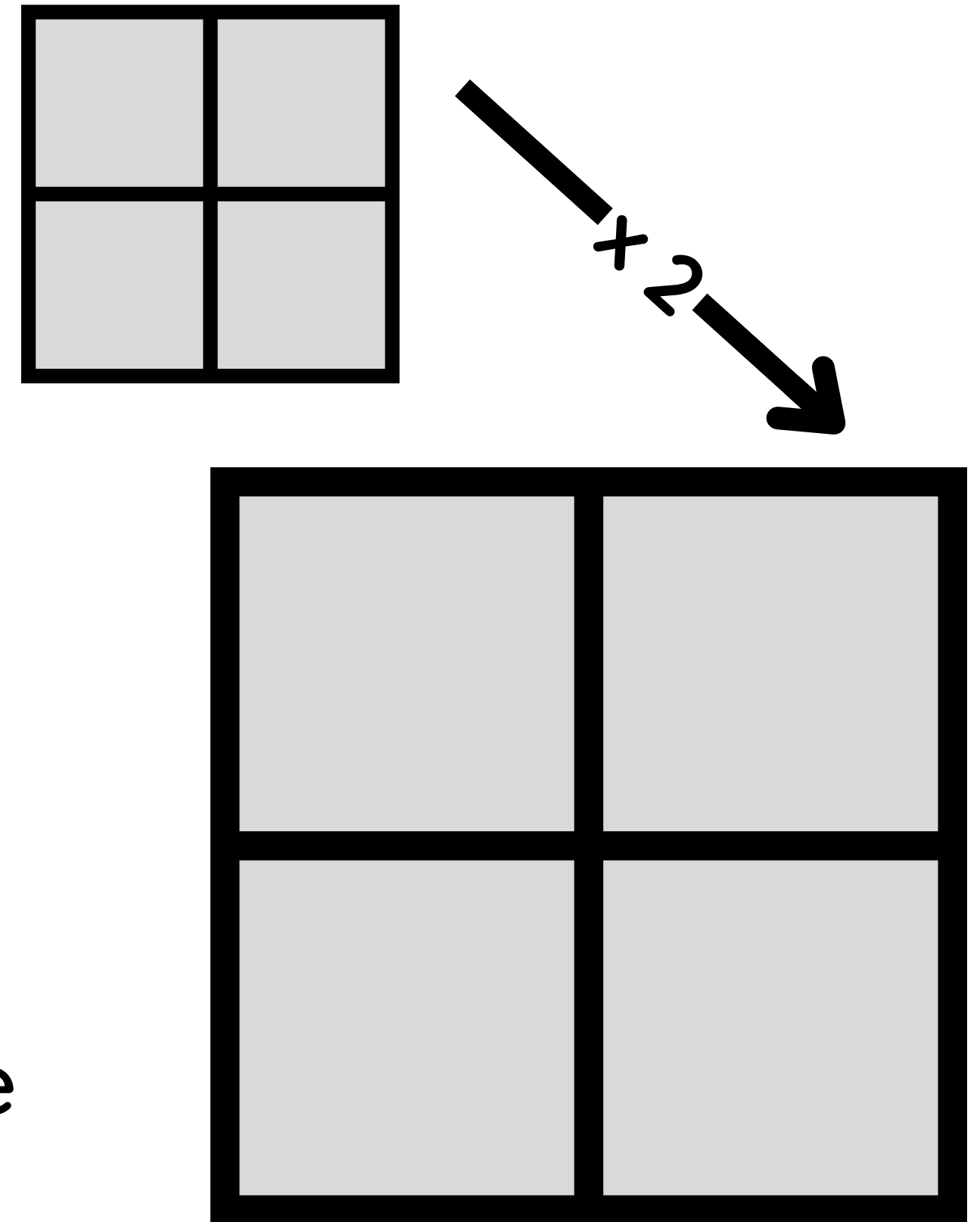
Every point in the image is the same distance from the center of rotation as their corresponding points in the pre-image.



DILATION

A rigid transformation in which a shape made bigger or smaller by a given factor called the scale factor.

A dilation that creates a larger image is called an enlargement. A dilation that creates a smaller image is called a reduction.



REAL LIFE APPLICATIONS

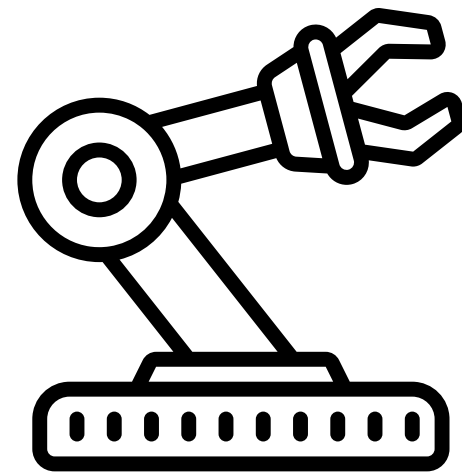
1

Animation and
computer graphics



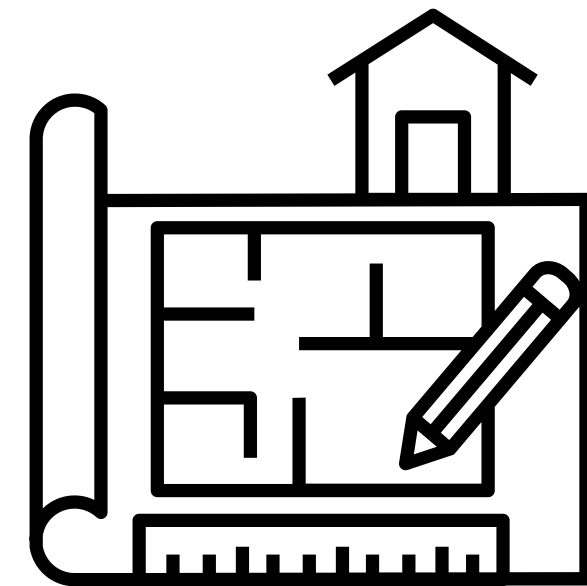
2

Robotics and
autonomous
vehicles



3

Architecture and
interior design



Thank you
for
listening!

